SEVENTH SEMESTER

Course Code	Course Name	Th	Tu	Pr	Credits	Evaluation			
						MSE	CA	ESE	Total
23CE731M	Advanced Data Analytics	3	-	-	3	15	15	70	100

Course Objectives	Course Outcomes
This course is intended to 1. Describe advanced data analytics workflows and architectures. 2. Apply preprocessing, visualization, and feature engineering techniques to datasets. 3. Analyze datasets using statistical and machine learning methods. 4. Develop predictive models for data-driven applications. 5. Examine emerging trends and ethical practices in analytics.	**Students will be able to** 1. Illustrate advanced data analytics workflows and applications. 2. Apply advanced preprocessing and visualization techniques on real-life datasets. 3. Experiment with various datasets using statistical and machine learning approaches. 4. Develop predictive analytical models for real-world applications. 5. Classify various emerging analytics technologies and intelligent decision systems.

Unit I: Advanced Analytics Foundations and Applications	**[08 Hrs.]**
Review of Data Analytics lifecycle and CRISP-DM methodology, Advanced analytics architecture and modern data ecosystem, Data Warehouses and Data Lakes, Predictive and Prescriptive Analytics concepts, Real-time and Stream Analytics overview, Data-driven decision systems, Analytics applications in healthcare, finance, manufacturing, education, and governance.	
Unit II: Data Preparation and Feature Engineering	**[10 Hrs.]**
Advanced data acquisition using APIs and streaming sources, Data integration and transformation techniques, Handling missing, noisy, and imbalanced datasets, Outlier detection methods, Feature selection and feature engineering, Dimensionality reduction concepts: PCA basics, Automated preprocessing pipelines using Pandas, NumPy, and Scikit-learn.	
Unit III: Statistical Analytics and Visual Intelligence	**[08 Hrs.]**
Inferential statistics and hypothesis testing basics, Correlation and regression analysis, Exploratory Data Analysis (EDA), Time-series analysis fundamentals, Advanced data visualization principles, Heatmaps and correlation visualization, Interactive dashboards and storytelling with data, Visualization using Matplotlib, Seaborn, and Power BI/Tableau.	
Unit IV: Deep Learning and Predictive Analytics	**[10 Hrs.]**
Advanced predictive analytics concepts, Introduction to Deep Learning for Data Analytics, Artificial Neural Networks (ANN), Deep Neural Networks (DNN), Fundamentals of Convolutional Neural Networks (CNN) and Recurrent Neural Networks (RNN), Auto encoders, Time-series and sequential data analytics, Model optimization and hyperparameter tuning basics, Model evaluation metrics: Precision, Recall, F1-score, ROC-AUC, Use of DL frameworks in scientific analysis of data	
Unit V: Emerging Trends and Multidisciplinary Analytics Workflows	**[09 Hrs.]**
Multidisciplinary applications of Data Analytics in healthcare, finance, education, and manufacturing, Analytics workflow: data acquisition, preprocessing, visualization, modeling, and decision support, Cloud and IoT-based analytics overview, Dashboard-driven analytics systems, Emerging trends in analytics: Generative AI, Explainable AI (XAI), Agentic AI, and AutoML concepts, Case studies: healthcare prediction, fraud detection, smart traffic analytics, predictive maintenance, and social media analytics	

Text Books/ Reference Books

S. N.	Title	Authors	Edition	Publisher
1	Data Science for Business	Foster Provost, Tom Fawcett	2nd	O'Reilly Media
2	Python for Data Analysis	Wes McKinney	3rd	O'Reilly Media
3	Introduction to Machine Learning with Python	Andreas Müller, Sarah Guido	2nd	O'Reilly Media
4	Data Mining: Concepts and Techniques	Jiawei Han, Micheline Kamber, Jian Pei	4th	Morgan Kaufmann

		JULY 2026	NEP 1.0	Applicable for 2026-27
Chairman - BoS	Dean – Academics	Date of Release	Version	